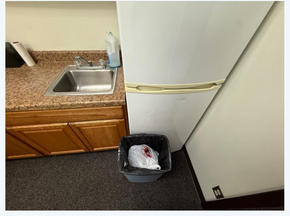
## Memory-conditioned Planning

You are a robot starting from the bedroom. Show me your plan to grab the milk from the fridge.

Answer: A. I walk to the fridge, open the fridge, grab the milk and close the fridge. B. I walk to the fridge, close the fridge, grab the milk and open the fridge. C. I walk to the fridge, grab the milk, open the fridge and close the fridge. D. I walk to the fridge, open the fridge, close the fridge and grab the milk

## Action Reasoning under Complex Env.



My next action is to open the lower fridge door. Based on the current state, how should I achieve this action?

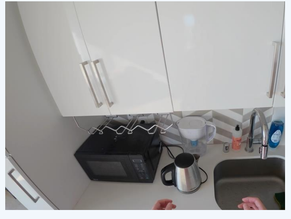
Answer: A. Open the lower fridge door. B. Move the trash can to the left. C. Empty the trash can D. Close the fridge door.

## Direction Recognition

In this video clip, what does the camera wearer do?

Answer: A. Walking straight B. Turning left C. Turning right D. Standing up E. Sitting down

## Next Action Prediction:



The abstract task is to bring kettle to bedroom. I walked to the kitchen counter. What will I do next?

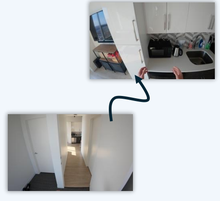
Answer: A. Pick up the kettle. B. Pick up the purifier. C. Switch on the kettle D. Go to the bedroom

## Action Recognition

In this video clip, describe the manipulation action: what is the camera wearer doing?

Answer: A. picking up the purifier B. Pouring the water C. Closing the lid D. boiling the paper

## Distance Estimation



In this video clip, how far does the subject walk (in meters)?

Answer: 3.2m